1.1.1Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum is calculated using the formula:
where:
 is the mass of the object (in kilograms).
 is the velocity of the object (in meters per second).

Code:

```
m=float(input(""))

v=float(input(""))

momentum=m*v

print(f"{momentum:.2f}kgm/s")
```

1.1.2. Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Code:

```
n=int(input(""))

if 0<n<10:

    print(n*n)

elif n<100:

    result=n**0.5

    print(f"{result:.2f}")

elif n<1000:

    result=round(n**(1/3),2)

    print(f"{result:.2f}")

else:

    print("Invalid")
```

1.1.3.Write a Python program to calculate number of days between two dates.

Code:

```
from datetime import date

from datetime import date

y1, m1, d1 = map(int, input().split('-'))

y2, m2, d2 = map(int, input().split('-'))

date1= date(y1,m1,d1)

date2= date(y2,m2,d2)

print((date2 - date1).days)
```

1.1.4. Write a Python program to reverse the digits of the number and print the reversed number.

Code:

```
n=int(input())

reverse=0

while n>0:

	digit=n%10

	reverse=reverse*10+digit

	n=n//10

print(reverse)
```

1.1.5. Write a Python program to reverse the digits of the number and print the reversed number.

Code:

```
n=int(input(""))

for i in range(1,11):

    print(n, "x", i, "=", n*i)
```

1.2.1. Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Code:

```
n=int(input(""))

marks= list(map(int ,input().split()))

if any(mark<40 for mark in marks):

    print("Fail")

else:

    aggregate = sum(marks)/n

    print("Aggregate Percentage:",f"{aggregate:.2f}")

    if aggregate>75:

        print("Grade: Distinction")

    elif 60<=aggregate<75:

        print("Grade: First Division")
```

```
    elif 60>aggregate>=50:
        print("Grade: Second Division")
    elif 50>aggregate>=40:
        print("Grade: Third Division")
```

1.2.2. Write a Python program that uses recursion to print the first **n** terms of the Fibonacci series.

Code:

```
def fibonacci(n):
    if n==1:
        return 0
    elif n==2:
        return 1
    else:
        return fibonacci(  n-1 )+ fibonacci(n-2)
n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
fibonacci(n)
```

1.2.3.Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Code:

```
n=int(input(""))
```

```
for i in range(1,n+1):
    print("* " * i)
```

1.2.4. Write a Python program to print a right-angled triangle pattern of numbers.

Code:

```
n=int(input(""))
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j, end=" "  )
print()
```